ICT 6535: Advanced Artificial Intelligence

Lecture 1 — Introduction to AI

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Course Contents

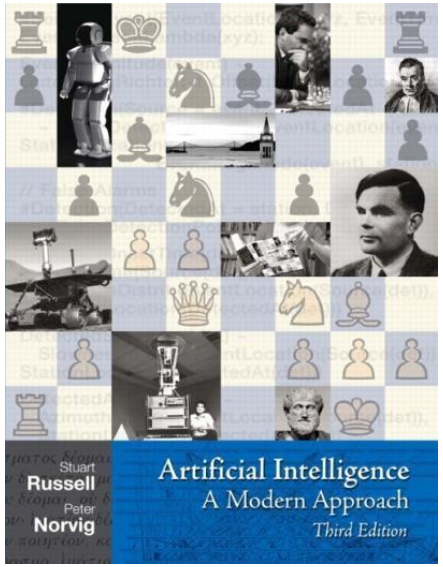
- 17.10.2026: Lecture 1 – Introduction
- 29.10.2026 (Extra): Lecture 2 – Agents in AI
- 31.10.2026: Lecture 3 – Uninformed Search Techniques
- 07.11.2026: Lecture 4 – Informed Search Techniques
- 14.11.2026: Midterm 1 & Lecture 5 – Adversarial Search
- 21.11.2026: Lecture 6 – Uncertainty and Utilities
- 28.11.2026: Lecture 7 – Markov Decision Processes
- 05.12.2026: No Lecture
- 12.12.2026: Midterm 2
- 02.01.2027: Lecture 8 – Reinforcement Learning
- 09.01.2027: Lecture 9 – Learning from Data
- 16.01.2027: Lecture 10 – Performance Improvement of ML Techniques
- 23.01.2027: Lecture 11 – Multi-Layer Neural Networks
- 30.01.2027: Lecture 12 – Revisions

Course Assessments

- Midterm Exam 1 (30%)
    - Tentative date: 14.11.2026
    - Syllabus: Lecture 1-4
- Midterm Exam 2 (30%)
    - Tentative date: 12.12.2026
    - Syllabus: Lecture 5-7
- Final Exam (40%)
    - Syllabus: To be decided

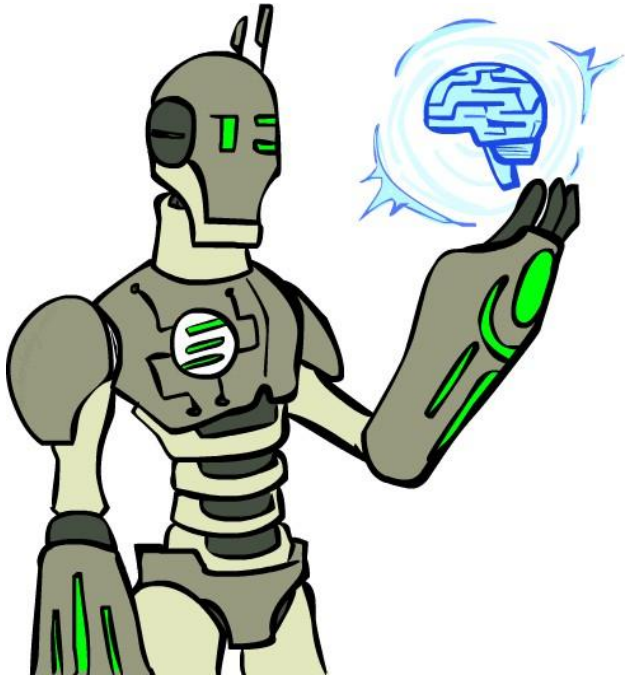
Reference Book

- Artificial Intelligence: A Modern Approach (3rd Ed)
  - Stuart Russell & Peter Norvig



Contents

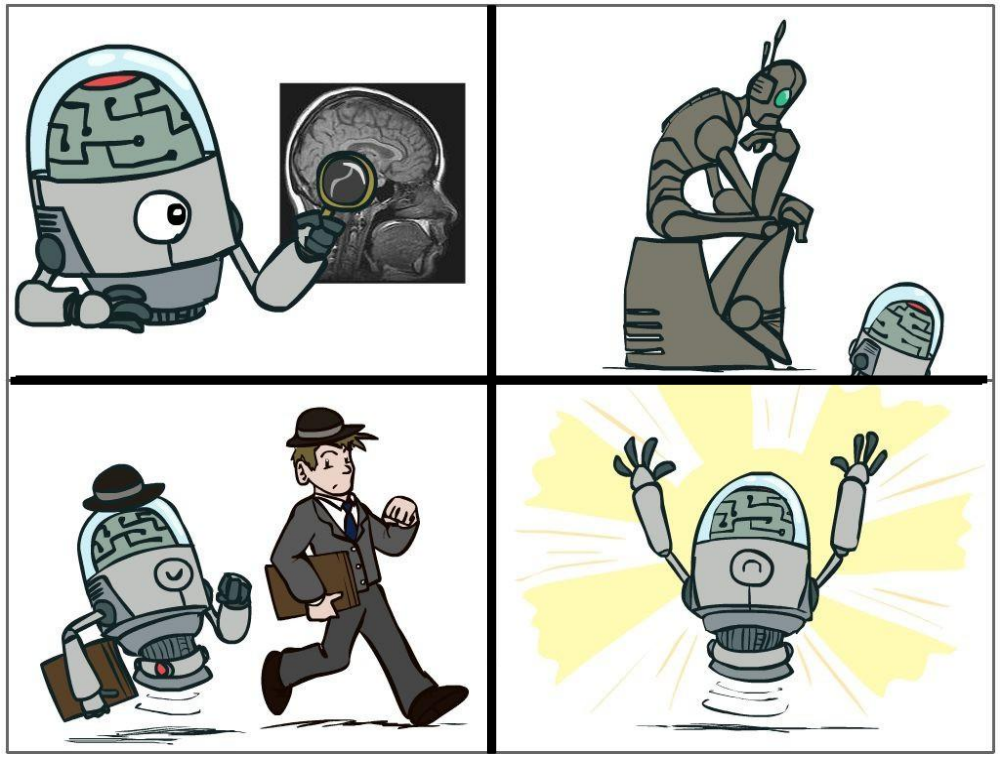
- What is artificial intelligence?
- What can AI do?
- What is this course?



Artificial Intelligence

Artificial Intelligence is the science of making machines that

Think like people



Think rationally

Act like people

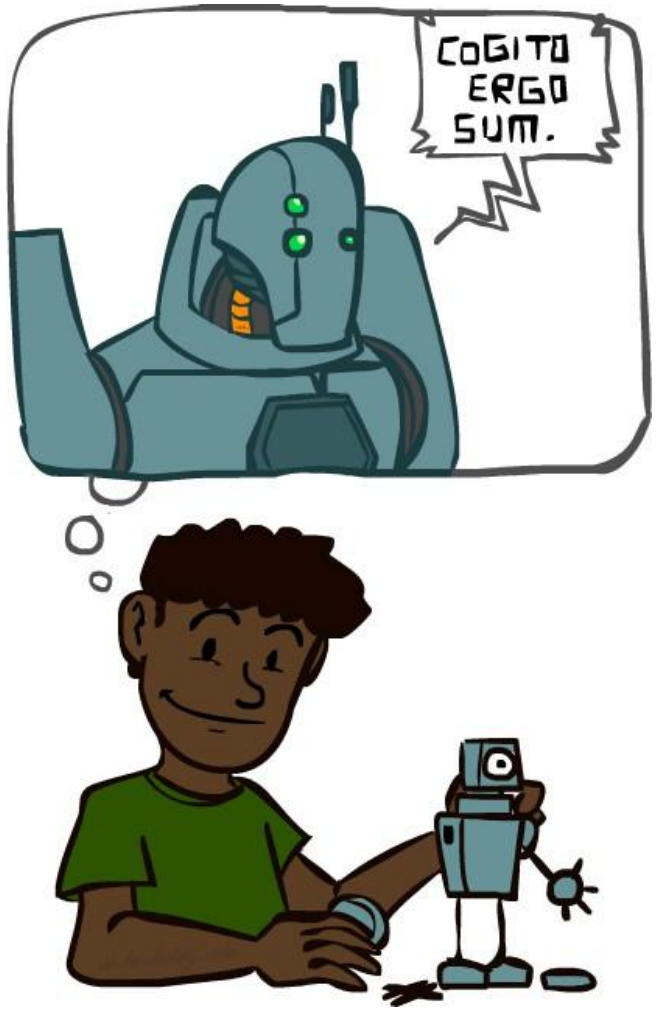
Act rationally

Rational Decisions

- **We'll use the term rational in a very specific, technical way:**
  - Rational: *maximally achieving pre-defined goals*
  - Goals are expressed in terms of the **utility** of outcomes
  - World is uncertain, so we'll use **expected** utility
  - Being rational means acting to **maximize your expected utility**

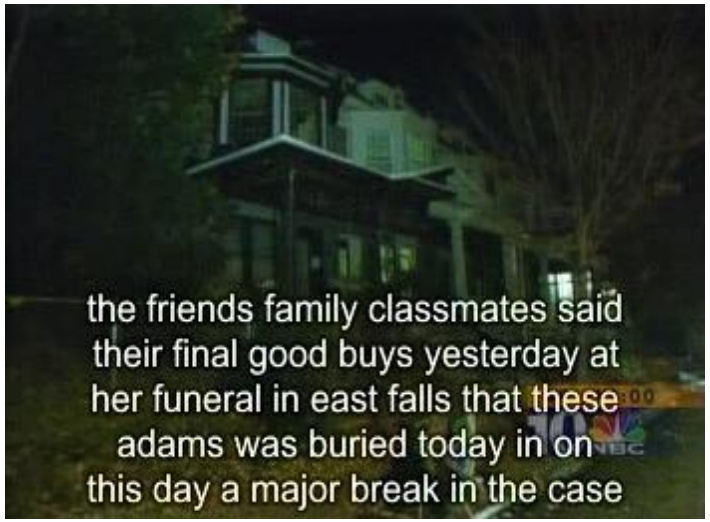
A (Short) History of AI

- 1940-1950: Early days: neural and computer science meet
  - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain
  - 1950: Turing's “Computing Machinery and Intelligence”
- 1950—70: Excitement! Logic-driven
  - 1950s: Early AI programs, including Samuel's checkers program,
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
  - 1969: Minsky & Papert: perceptrons can’t learn XOR/parity!
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
  - 1988—93: Expert systems industry busts: “AI Winter”
- 1990—2010: Statistical approaches, agents
  - Resurgence of probability, focus on uncertainty
  - Agents and learning systems… “AI Spring”?
  - 1992: TD-Gammon achieves human-level play at backgammon
  - 1997: Deep Blue defeats Gary Kasparov at chess
  - 2002: Embodied AI; Roomba vacuum invented
- 2010—Present: Big Data, GPUs, Deep Learning



Natural Language Processing

- Speech technologies (e.g. Siri)
  - Automatic speech recognition (ASR)
  - Text-to-speech synthesis (TTS)
  - Dialog systems
- Language processing technologies
  - Question answering
  - Machine translation



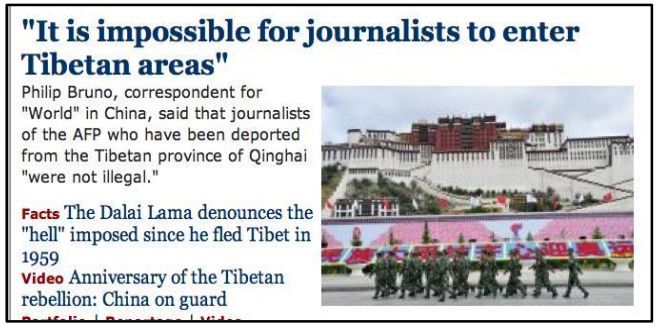


"Il est impossible aux journalistes de rentrer dans les régions tibétaines"

Bruno Philip, correspondant du "Monde" en Chine, estime que les journalistes de l'AFP qui ont été expulsés de la province tibétaine du Qinghai "n'étaient pas dans l'illégalité".

Les faits Le dalaï-lama dénonce l'"enfer" imposé au Tibet depuis sa fuite, en 1959

Vidéo Anniversaire de la rébellion



"It is impossible for journalists to enter Tibetan areas"

Philip Bruno, correspondent for "World" in China, said that journalists of the AFP who have been deported from the Tibetan province of Qinghai "were not illegal."

Facts The Dalai Lama denounces the "hell" imposed since he fled Tibet in 1959

Video Anniversary of the Tibetan rebellion: China on guard

- Web search
- Text classification, spam filtering, etc…

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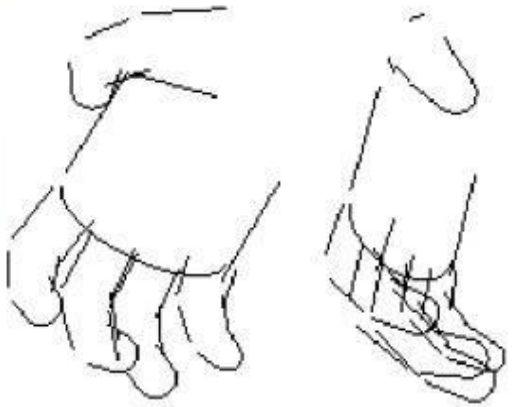
Vision (Perception)

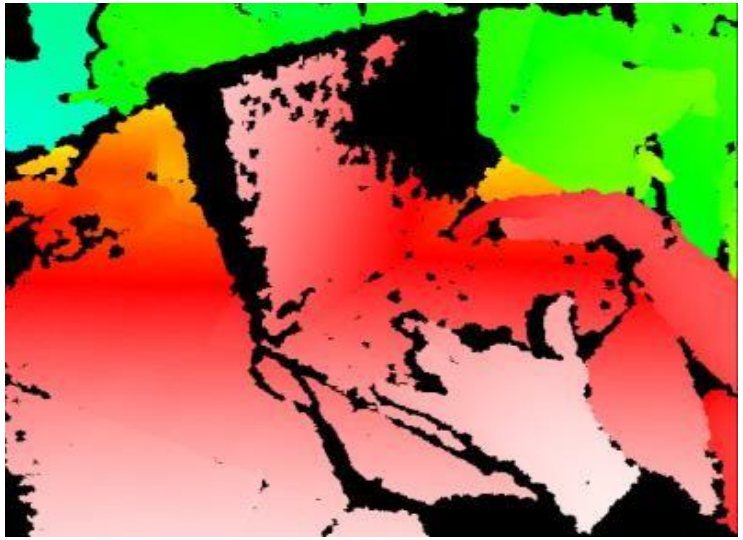
- Object and face recognition
- Scene segmentation
- Image classification











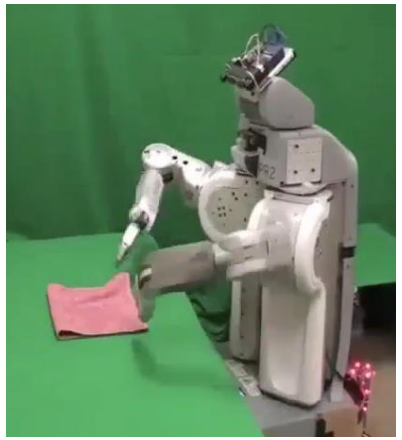
Images from Erik Sudderth (left), wikipedia (right)

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Robotics

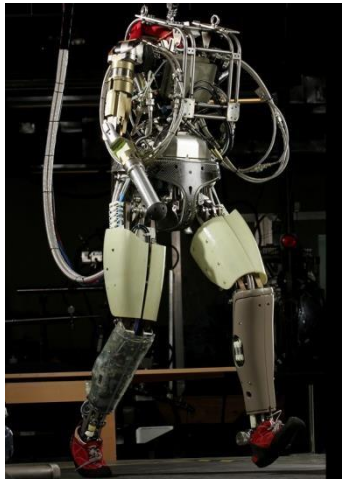
- Robotics
  - Part mech. eng.
  - Part AI
  - Reality much harder than simulations!

- Technologies
  - Vehicles
  - Rescue
  - Soccer!
  - Lots of automation…









Decision Making

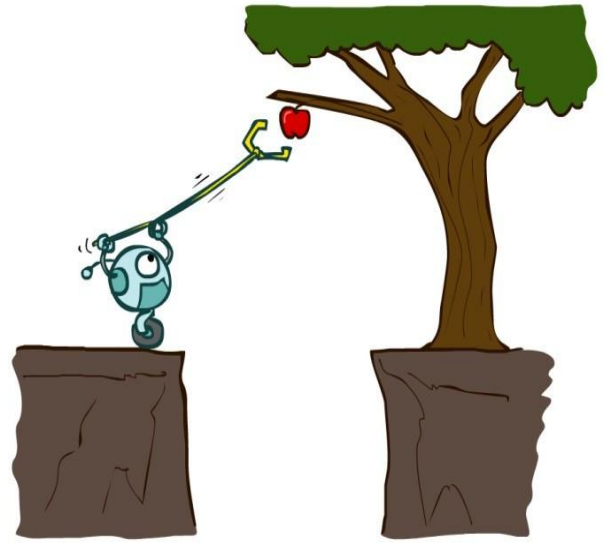
- **Applied AI involves many kinds of automation**
  - Scheduling, e.g. airline routing, military
  - Route planning, e.g. Google maps
  - Medical diagnosis
  - Web search engines
  - Spam classifiers
  - Automated help desks
  - Fraud detection
  - Product recommendations
  - … Lots more!

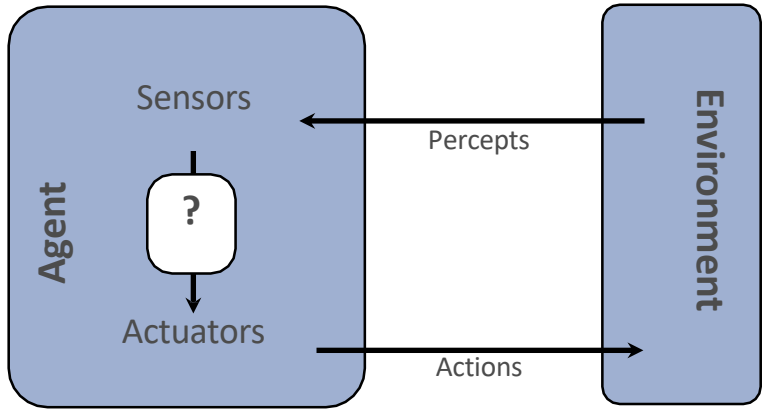




About This Course

- **An agent is an entity that perceives and acts.**
- **A rational agent selects actions that maximize its (expected) utility.**
- **Characteristics of the percepts, environment, and action space dictate techniques for selecting rational actions**
- **This course is about:**
  - General AI techniques for a variety of problem types







Thank you for your attention!

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